

Lab Records

Logic building Question

Question 1:

Given a number n , check whether it is even or odd. Return **true** for even and **false** for odd.

Examples:

Input: $n = 15$

Output: *false*

Explanation: $15 \% 2 = 1$, so 15 is odd .

Input: $n = 44$

Output: *true*

Explanation: $44 \% 2 = 0$, so 44 is even.

Question 2:

Given a number n , we need to print its table.

Examples :

Input: 5

Output:

$5 * 1 = 5$

$5 * 2 = 10$

$5 * 3 = 15$

$5 * 4 = 20$

$5 * 5 = 25$

$5 * 6 = 30$

$5 * 7 = 35$

$5 * 8 = 40$

$5 * 9 = 45$

$5 * 10 = 50$

Question 3:

Given a positive integer n , we have to find the sum of squares of first n natural numbers.

Examples :

Input : $n = 2$

Output: 5

Explanation: $1^2 + 2^2 = 5$

Input : $n = 8$

Output: 204

Explanation : $1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 + 7^2 + 8^2 = 204$

Question 4:

Given an Integer n , find the reverse of its digits.

Examples:

Input: $n = 122$

Output: 221

Explanation: By reversing the digits of number, number will change into 221.

Input: $n = 200$

Output: 2

Explanation: By reversing the digits of number, number will change into 2.

Question 5:

Given a number n , find the sum of its digits.

Examples :

Input: $n = 687$

Output: 21

Explanation: The sum of its digits are: $6 + 8 + 7 = 21$

Input: $n = 12$

Output: 3

Explanation: The sum of its digits are: $1 + 2 = 3$

Question 6:

Given a number n , find the count of its digits.

Examples :

Input: $n = 687$

Output: 3

Explanation: The number 687 has 3 digits (6, 8, 7).

Input: $n = 12$

Output: 2

Explanation: The number 12 has 2 digits (1, 2).

Input: $n = 12345$

Output: 5

Explanation: The number 12345 has 5 digits (1, 2, 3, 4, 5).

Question 7:

Given a positive integer, check if the number is prime or not. A prime is a natural number greater than 1 that has no positive divisors other than 1 and itself. Examples of the first few prime numbers are {2, 3, 5, ...}

Examples :

Input: $n = 11$

Output: true

Input: $n = 15$

Output: false

Input: $n = 1$

Output: false

Question 8:

Given three sides, check whether triangle is valid or not.

Examples:

Input : $a = 7, b = 10, c = 5$

Output : Valid

We can draw a triangle with the given three edge lengths.

Input : $a = 1, b = 10, c = 12$

Output : Invalid

We can not draw a triangle with the given three edge lengths.

Question 9:

Given the number n ($n \geq 0$), find a factorial. Factorial of n is defined as $1 \times 2 \times \dots \times n$. For $n = 0$, factorial is 1. We are going to discuss iterative and recursive programs in this post.

Examples:

Input: $n = 5$

Output: 120

Explanation: $5! = 5 * 4 * 3 * 2 * 1 = 120$

Input: $n = 4$

Output: 24

Explanation: $4! = 4 * 3 * 2 * 1 = 24$

Input: $n = 0$

Output: 1

Input: $n = 1$

Output: 1

Question 10:

Given two positive integers **a** and **b**, the task is to find the GCD of the two numbers.

Note: The GCD (Greatest Common Divisor) or HCF (Highest Common Factor) of two numbers is the largest number that divides both of them.

Examples:

Input: $a = 20, b = 28$

Output: 4

Input: $a = 60, b = 36$

Output: 12

Question 11:

Given two positive integers **a** and **b**. Find the Least Common Multiple (LCM) of **a** and **b**.

LCM of two numbers is the smallest number which can be divided by both numbers.

Input : $a = 10, b = 5$

Output : 10

Explanation : 10 is the smallest number divisible by both 10 and 5

Input : $a = 5, b = 11$

Output : 55

Explanation : 55 is the smallest number divisible by both 5 and 11

Question 12:

Given a positive integer **n**, the task is to print the Fibonacci series up to **n terms**. The Fibonacci sequence is a series where the next term is the sum of the previous two terms. The first two terms of the Fibonacci sequence are **0** followed by **1**. The Fibonacci sequence: 0, 1, 1, 2, 3, 5, 8, 13, 21 ...

Example 1:

Input: n = 5

Output: 0 1 1 2 3

Explanation: The first 5 terms of the Fibonacci series are printed.

Example 2:

Input: n = 8

Output: 0 1 1 2 3 5 8 13

Explanation: The first 8 terms of the Fibonacci series are printed.

Question 13:

Given a non negative number **n**, the task is to convert the given number into an equivalent **binary representation**.

Examples:

Input: n = 12

Output: "1100"

Input: n = 33

Output: "100001"

Question 14:

Given a binary number **n**, the task is to convert the given number into its equivalent decimal representation.

The decimal number system has base 10, while the binary number system has base 2.

Example 1:

Input: n = 1100

Output: 12

Example 2:

Input: n = 100001

Output: 33

Question 15:

Given an integer **n**, find whether the number is Palindrome or not. A number is a Palindrome if it remains the same when its digits are reversed.

Examples:

Input: $n = 12321$

Output: Yes

Explanation: 12321 is a Palindrome number because after reversing its digits, the number becomes 12321 which is the same as the original number.

Input: $n = 1234$

Output: No

Explanation: 1234 is not a Palindrome number because after reversing its digits, the number becomes 4321 which is different from the original number.

Time and Space

Q1:

```
for (int i = 0; i < n; i += 2) {  
    printf("%d ", i);  
}
```

- **Hint:** Linear loop with constant increment.
- **Time Complexity:** ?
- **Space Complexity:** ?

Q2:

```
for (int i = n; i > 0; i = i / 2) {  
    printf("%d ", i);  
}
```

- **Hint:** Divide-by-constant loop
- **Time Complexity:** ?
- **Space Complexity:** ?

Q3:

```
void fun(int n) {  
    if (n == 0) return;  
    fun(n / 2);  
    printf("%d ", n);  
}
```

- **Hint:** One recursive call per halving
- **Time Complexity:** ?

- **Space Complexity:** ?

Q4:

```
int sum(int n) {
    if (n == 0) return 0;
    return n + sum(n - 1);
}
```

- **Hint:** n recursive calls
- **Time Complexity:** ?
- **Space Complexity:** ?

Q5:

```
void expFunc(int n) {
    if (n == 0) return;
    expFunc(n - 1);
    expFunc(n - 1);
}
```

- **Hint:** Two calls per level → binary tree
- **Time Complexity:** ?
- **Space Complexity:** ?

Q6:

```
for (int i = 0; i < n; i++) {
    for (int j = 1; j < n; j = j * 2) {
        printf("%d %d\n", i, j);
    }
}
```

- **Hint:** Outer = n, Inner = log n
- **Time Complexity:** ?
- **Space Complexity:** ?

Q7:

```
for (int i = 0; i < n; i++) {
    if (i % 2 == 0)
        printf("%d ", i);
}
```

- **Hint:** Runs n times, condition doesn't affect complexity
- **Time Complexity:** ?
- **Space Complexity:** ?

Q8:

```
for (int i = 0; i < n; i++) {
    int a = i * i;
    printf("%d ", a);
}
```

- **Hint:** One small variable per loop
- **Time Complexity:** ?
- **Space Complexity:** ?

Q9:

```
for (int i = 0; i < n; i++) {
    printf("Linear %d\n", i);
}
for (int i = 1; i < n; i *= 2) {
    printf("Log %d\n", i);
}
int a = 5 + 3;
```

- **Time Complexity:** ?
- **Space Complexity:** ?

Q10:

```
void loopThenRecurse(int n) {
    for (int i = 0; i < 10; i++)
        printf("%d ", i);
    if (n == 0) return;
    loopThenRecurse(n - 1);
}
```

- **Hint:** Constant loop, n recursive calls
- **Time Complexity:** ?
- **Space Complexity:** ?

Array Questions

Question 1:

Given an array of integers, calculate the average of its elements.

Examples:

Input: arr = [2, 4, 6, 8]

Output: 5

Explanation: $(2 + 4 + 6 + 8) / 4 = 20 / 4 = 5$.

Input: arr = [5, 15]

Output: 10

Explanation: $(5 + 15) / 2 = 20 / 2 = 10$.

Question 2:

Find the smallest element in an array.

Examples:

Input: arr = [4, 2, 7, 1, 9]

Output: 1

Explanation: The minimum element is 1.

Input: arr = [5, 5, 5]

Output: 5

Question 3:

Given an array, copy its elements into another array.

Examples:

Input: arr = [5, 10, 15]

Output: newArr = [5, 10, 15]

Explanation: All elements copied to new array.

Input: arr = [1]

Output: newArr = [1]

Question 4:

Check if a given array is a palindrome (same forward and backward).

Examples:

Input: arr = [1, 2, 3, 2, 1]

Output: Yes

Explanation: The array is same from front and back.

Input: arr = [1, 2, 3]

Output: No

Explanation: Forward = [1, 2, 3], Backward = [3, 2, 1]. Not equal.

Question 5:

Find the missing number in a given range from 1 to n.

Examples:

Input: arr = [1, 2, 4, 5], n = 5

Output: 3

Explanation: Range is 1 to 5, missing number is 3.

Input: arr = [2, 3, 1, 5], n = 5

Output: 4

Question 6:

Search an element in an array using linear search.

Examples:

Input: arr = [5, 3, 8, 6], target = 8

Output: Found at index 2

Explanation: $\text{arr}[2] = 8$ matches target.

Input: $\text{arr} = [1, 2, 3]$, $\text{target} = 4$

Output: Not Found

Question 7:

Left rotate the array by one position.

Examples:

Input: $\text{arr} = [1, 2, 3, 4, 5]$

Output: $[2, 3, 4, 5, 1]$

Input: $\text{arr} = [7, 8]$

Output: $[8, 7]$

Question 8:

Find the difference between the largest and smallest element in an array.

Examples:

Input: $\text{arr} = [10, 4, 7, 15]$

Output: 11

Explanation: Largest = 15, Smallest = 4, Difference = $15 - 4 = 11$.

Input: $\text{arr} = [2, 2, 2]$

Output: 0

Question 9:

Find the second largest element in an array.

Examples:

Input: $\text{arr} = [10, 20, 4, 45, 99]$

Output: 45

Explanation: Largest = 99, Second largest = 45.

Input: $\text{arr} = [1, 2]$

Output: 1

Explanation: Largest = 2, second largest = 1.

Question 10:

Move all zeroes to the end of the array while maintaining the order of non-zero elements.

Examples:

Input: $\text{arr} = [0, 1, 0, 3, 12]$

Output: $[1, 3, 12, 0, 0]$

Input: $\text{arr} = [1, 0, 2, 0]$

Output: $[1, 2, 0, 0]$

//string practice

Question 1:

Given a string, count the number of vowels (a, e, i, o, u).

Examples:

Input: "hello"

Output: 2

Input: "education"

Output: 5

Question 2:

Reverse a given string without using library functions.

Examples:

Input: "world"

Output: "dlrow"

Input: "python"

Output: "nohtyp"

Question 3:

Check whether a given string is a palindrome or not.

(A string is palindrome if it reads the same backward as forward.)

Examples:

Input: "madam"

Output: Palindrome

Input: "hello"

Output: Not Palindrome

Question 4:

Count the number of words in a given string.

Examples:

Input: "I love programming"

Output: 3

Input: "ChatGPT is awesome"

Output: 3

Question 5:

Find the frequency of each character in a string.

Examples:

Input: "apple"

Output:

```
a → 1  
p → 2  
l → 1  
e → 1
```

Input: "banana"

Output:

```
b → 1  
a → 3  
n → 2
```

Question 6:

Convert all lowercase letters in a string to uppercase and vice versa.

Examples:

Input: "HeLLo"

Output: "hEllO"

Input: "Python"

Output: "pYTHON"

Question 7:

Find the length of a string without using the built-in `strlen()` function.

Examples:

Input: "OpenAI"

Output: 6

Input: "C programming"

Output: 13

Question 8:

Remove all spaces from a given string.

Examples:

Input: "Hello World"

Output: "HelloWorld"

Input: "Data Structures"

Output: "DataStructures"

Question 9:

Count the number of digits, alphabets, and special characters in a string.

Examples:

Input: "A1@B2#C3"

Output:

Alphabets = 3

Digits = 3

Special Characters = 2

Question 10:

Find the first non-repeating character in a string.

Examples:

Input: "swiss"

Output: "w"

Input: "success"

Output: "u"
